

# Service Layers Explained

Coordinating Business Logic in Apex

FFLib Series · Session 002

Andrew Fawcett

Code With Sally

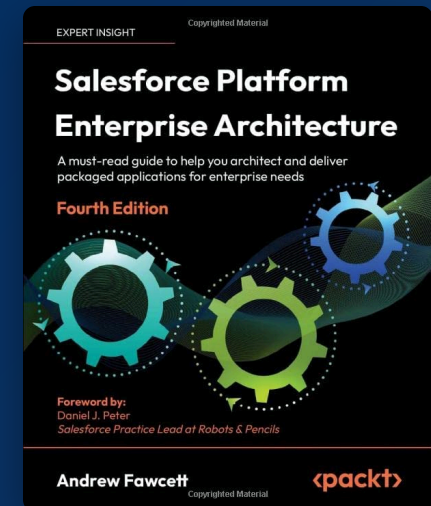


# About me

Independent Salesforce Consultant | Former CPO,  
Salesforce, Heroku | CTO, FinancialForce.com

- **Independent consultant and architect** — Salesforce platform development, PaaS integrations, and agentic AI
- **Open source and guidance** — creator of DLRS, Apex frameworks, and *Salesforce Platform Enterprise Architecture*
- **Salesforce Customers and Partners Advisory** — architecture and scale, AI-enhanced workflows, org health and codebase mentoring, and ISV product advisory

Reduce complexity, unblock engineering, and leave architectures — and teams — able to evolve.



# The series

## SESSION



Session #1 - Separation of Concerns in Apex: Why Your Future Self Will Thank You



Session #2 - Service Layers Explained: Coordinating Business Logic in Apex



Session #3 - Domain vs Service: Where Should Your Apex Logic Live?



Session #4 - Query Logic as a First-Class Architecture Concern

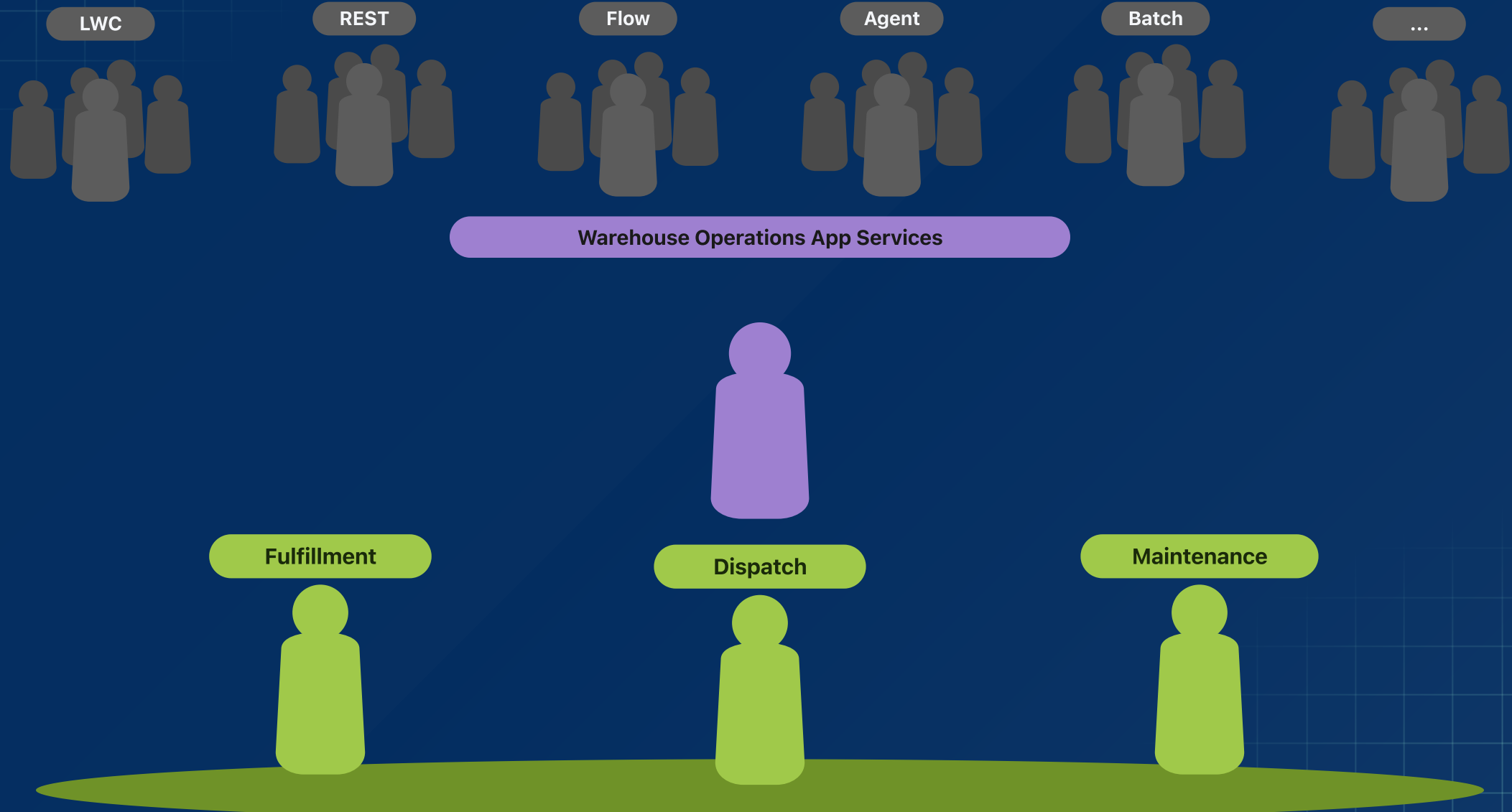


Session #5 - Mocking in Apex: Why It Changes Everything



Session #6 - Enterprise-Scale Apex Across Multiple Packages

# Coordinating Business Logic in Apex





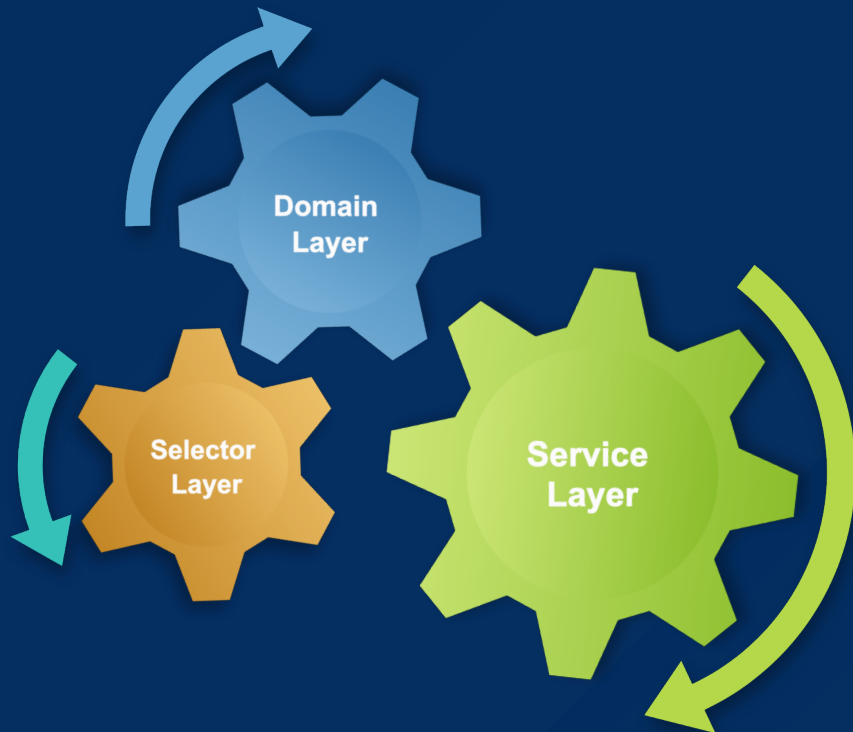
# Service Layer

“ *Defines an application's boundary with a layer of services that establishes a set of available operations and coordinates the application's response in each operation.* ”

[Martin Fowler](#) — *Patterns of Enterprise Application Architecture*



# Service Layers Explained



## SERVICE LAYERS EXPLAINED

Separation of Concerns Recap ☐

The Unit of Work ☐

Service Layer Principles ☐

Warehouse Operations App Overview ☐

Warehouse Operations App Code ☐

Warehouse Operations Agent ☐

# Salesforce Platform Layers

## Presentation

- **You define:** Layouts, Custom Buttons, Flow, Record Types, Formula, Reports, Dashboards, Agent Builder
- **You code:** Apex Controllers, Visualforce, Aura, Lightning Components, React

## Integration

- **You define:** Custom Objects, Custom Fields
- **You code:** Apex SOAP, Apex REST, Invocable Methods, Agent Actions, MCP, `global` Apex

## Business Logic

- **You define:** Formula, Validation Rules, Workflow, Process Builder, Flow, Sharing Rules, Prompt Builder
- **You code:** Apex Services, DataWeave, Callouts

## Data Access

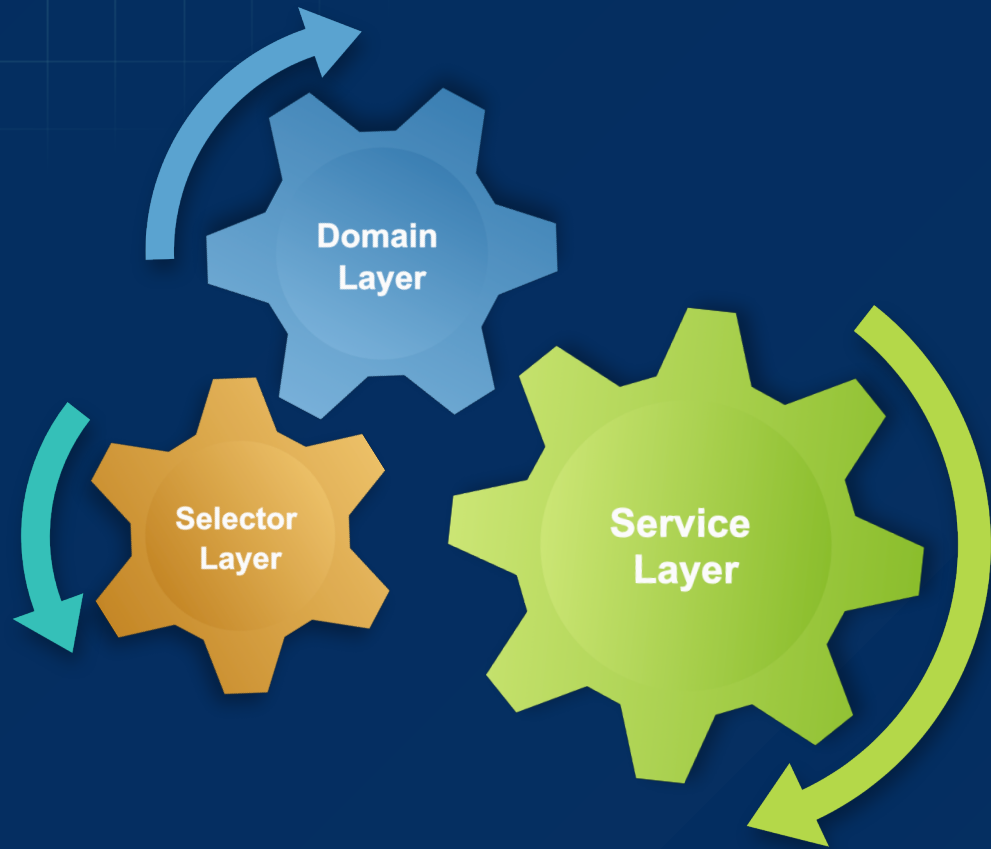
- **You define:** Data Loaders, Permission Sets, Profiles
- **You code:** SOQL, SOSL and Salesforce APIs

## Database

- **You define:** Custom Objects, Custom Fields, Relationships, Rollups, D360
- **You code:** Apex Triggers, External Data Sources (External Objects)



# Apex Enterprise Patterns (fflib) - Color Chart



-  Client
-  Service
-  Domain
-  Trigger Handler
-  Selector

# Suggested Class Folder Layout

## Presentation

■ FulfillmentReleaseController.cls   ■ WarehouseDispatchController.cls   ■ MaintenanceCompleteController.cls   .

*// /classes/controllers*

## Integration

■ ReleaseOrders.cls   ■ DispatchWarehouse.cls   ■ FulfillmentResource.cls

*// /classes/actions · /classes/restapis*



## Business Logic

■ FulfillmentService.cls   ■ DispatchService.cls   ■ MaintenanceService.cls

*// /classes/services – named for the process*

■ Robots.cls   ■ FulfillmentOrders.cls   ■ FulfillmentLines.cls   ...

*// /classes/domains – named for the object*

## Data Access

■ RobotsSelector.cls   ■ FulfillmentLinesSelector.cls   ■ WarehousesSelector.cls

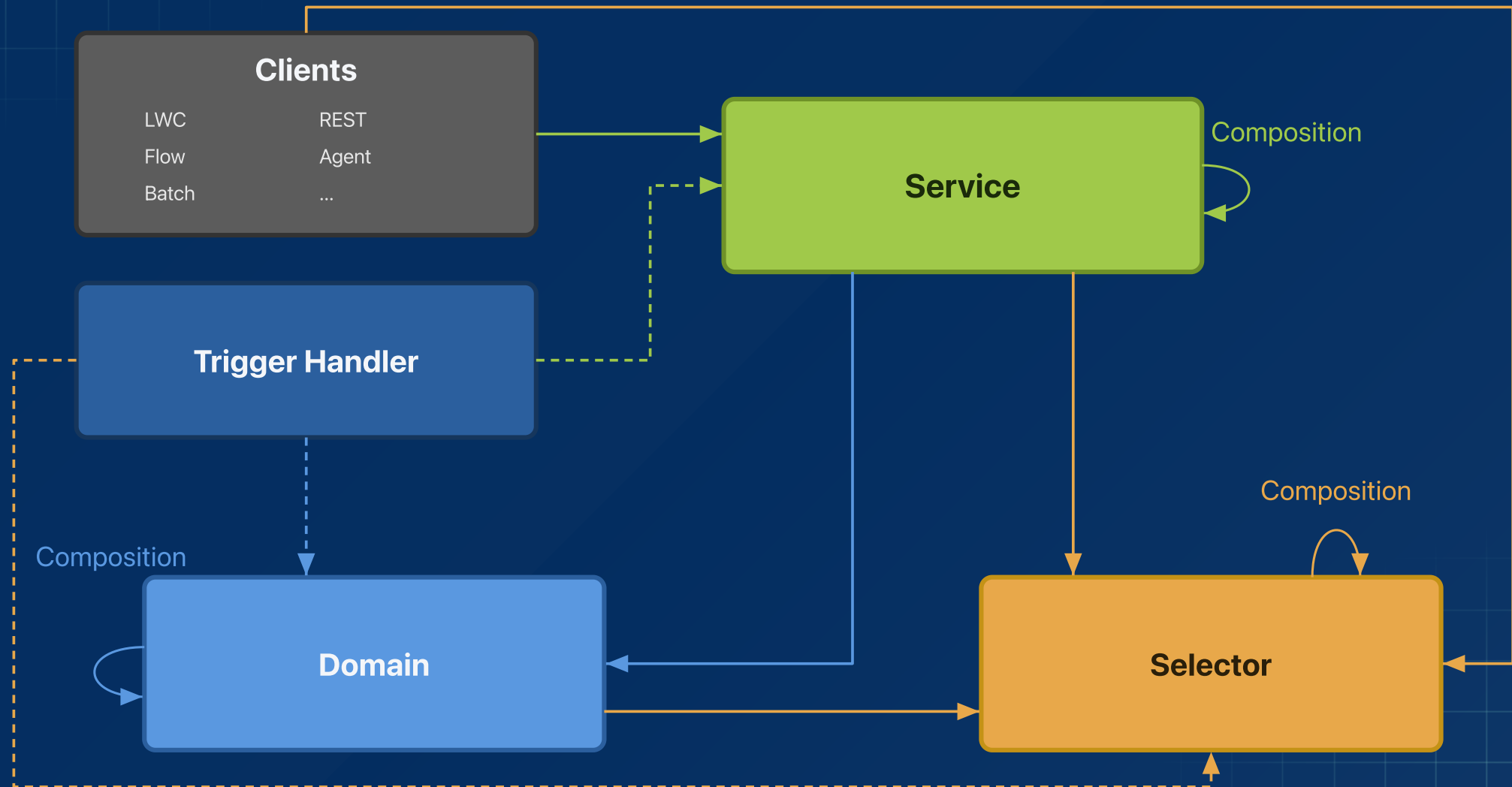
*// /classes/selectors*

## Database

■ RobotsTriggerHandler.cls

*// /classes/triggerHandlers*

# Who can call whom



# Who can call whom

| CALLER                                             | Service | Domain | Selector |
|----------------------------------------------------|---------|--------|----------|
| <b>Client</b> (LWC, REST, Flow, Agent, Batch, ...) | •       |        | •        |
| <b>Trigger Handler</b>                             | •       | •      | •        |
| <b>Service</b>                                     | •       | •      | •        |
| <b>Domain</b>                                      |         | •      | •        |
| <b>Selector</b>                                    |         |        | •        |

# How durable is your code when change occurs?

How users interact with your application changes a lot — protect against it with SoC



🤔 Business Logic Impact

WarehouseDispatchController.cls DispatchWarehouse.cls FulfillmentReleaseController.cls ReleaseOrders.cls

```
// logic copied — every new client type
VF · Aura · LWC · Agent Action · React · Next?
└─ WarehouseDispatchController.dispatchWarehouse(...)
└─ DispatchWarehouse.execute(...)
└─ FulfillmentReleaseController.releaseOrder(...)
└─ ReleaseOrders.execute(...)
```

① Claudeforce. Salesforce + Anthropic, August 2026 — yet another interaction layer on your application.



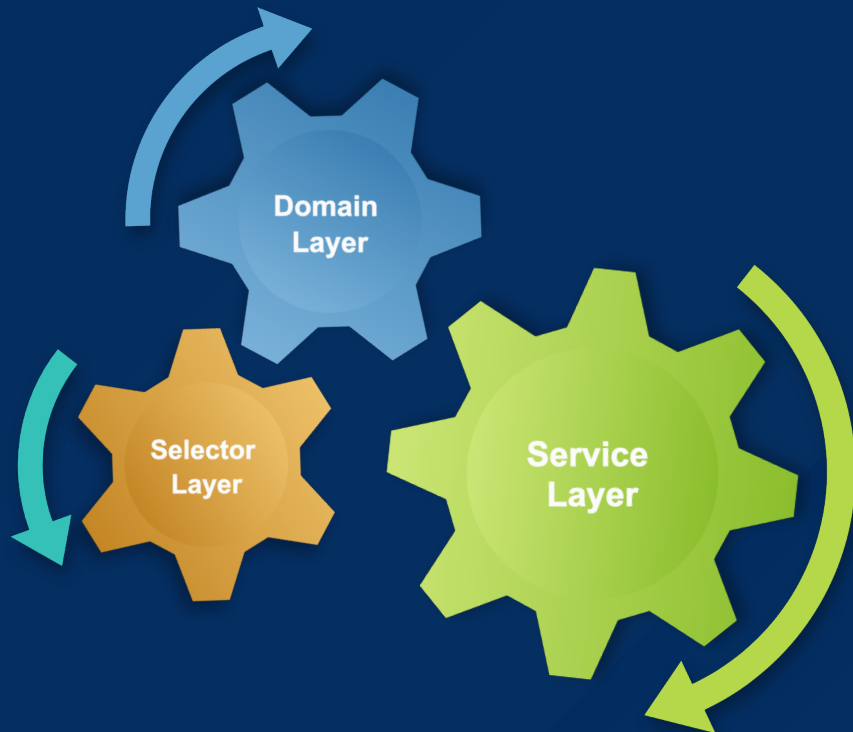
DispatchService.cls FulfillmentService.cls

```
// same services — every client type
LWC · REST · Batch · Flow · Agent Action · Next?
└─ DispatchService.dispatchWarehouses(...)
└─ FulfillmentService.releaseOrders(...)
```

❤️ Your business logic is **the** most important logic in your app — and when it's everywhere, it becomes a **significant investment risk**



# Service Layers Explained



## SERVICE LAYERS EXPLAINED

Separation of Concerns Recap



The Unit of Work



Service Layer Principles



Warehouse Operations App Overview



Warehouse Operations App Code



Warehouse Operations Agent



# Unit of Work

- [Martin Fowler](#) — *Patterns of Enterprise Application Architecture*
  - “Maintains a list of objects affected by a business transaction and coordinates the writing out of changes and the resolution of concurrency problems.”
- One transaction: work completes or is rolled back explicitly
- Best practices: automatically bulkifies, runs in user mode
- Relationships: built in memory without having to insert parent
- `commitWork()` takes a savepoint and rolls back if anything fails
- In Apex that means: register work now, `commitWork()` once — see [AndyInTheCloud, June 2013](#)

# Without Unit Of Work — Lines: 73

## ■ Opportunity setup — raw DML

```
1 List<Opportunity> opps = new List<Opportunity>();
2 List<List<Product2>> productsByOpp = new List<List<Product2>>();
3 List<List<PricebookEntry>> pbesByOpp = new List<List<PricebookEntry>>();
4 List<List<OpportunityLineItem>> linesByOpp = new List<List<OpportunityLineItem>>();
5 for (Integer o = 0; o < 10; o++) {
6     Opportunity opp = new Opportunity();
7     // ... set Name, StageName, CloseDate
8     opps.add(opp);
9     List<Product2> products = new List<Product2>();
10    List<PricebookEntry> pbes = new List<PricebookEntry>();
11    List<OpportunityLineItem> lines = new List<OpportunityLineItem>();
12    for (Integer i = 0; i < o + 1; i++) {
13        Product2 product = new Product2();
14        // ... set Name
15        products.add(product);
16        PricebookEntry pbe = new PricebookEntry();
17        // ... set UnitPrice, IsActive, Pricebook2Id
18        pbes.add(pbe);
19        OpportunityLineItem oli = new OpportunityLineItem();
20        // ... set Quantity, TotalPrice
21        lines.add(oli);
22    }
23    productsByOpp.add(products);
24    pbesByOpp.add(pbes);
25    linesByOpp.add(lines);
26 }
27 insert opps;
28 List<Product2> allProducts = new List<Product2>();
29 for (List<Product2> products : productsByOpp) {
30     allProducts.addAll(products);
31 }
32 insert allProducts;
33 Integer oppIdx = 0;
34 List<PricebookEntry> allPbes = new List<PricebookEntry>();
35 for (List<PricebookEntry> pbes : pbesByOpp) {
36     List<Product2> products = productsByOpp[oppIdx++];
37     Integer lineIdx = 0;
38     for (PricebookEntry pbe : pbes) {
39         pbe.Product2Id = products[lineIdx++].Id;
40     }
41     allPbes.addAll(pbes);
42 }
43 insert allPbes;
44 oppIdx = 0;
45 List<OpportunityLineItem> allLines = new List<OpportunityLineItem>();
46 for (List<OpportunityLineItem> lines : linesByOpp) {
47     List<PricebookEntry> pbes = pbesByOpp[oppIdx];
48     Integer lineIdx = 0;
49     for (OpportunityLineItem oli : lines) {
50         oli.OpportunityId = opps[oppIdx].Id;
51         oli.PricebookEntryId = pbes[lineIdx++].Id;
52     }
53     allLines.addAll(lines);
54     oppIdx++;
55 }
56 insert allLines;
```

## ■ Zoom · lines 1–4

```
1 List<Opportunity> opps = new List<Opportunity>();
2 List<List<Product2>> productsByOpp = new List<List<Product2>>();
3 List<List<PricebookEntry>> pbesByOpp = new List<List<PricebookEntry>>();
4 List<List<OpportunityLineItem>> linesByOpp = new List<List<OpportunityLineItem>>();
```

## ■ Zoom · lines 47–56

```
47 for (List<PricebookEntry> pbes : pbesByOpp) {
48     List<Product2> products = productsByOpp[oppIdx++];
49     Integer lineIdx = 0;
50     for (PricebookEntry pbe : pbes) {
51         pbe.Product2Id = products[lineIdx++].Id;
52     }
53     allPbes.addAll(pbes);
54 }
55 }
```

## ■ Zoom · lines 36, 43, 57, 73

```
36 insert opps;
43 insert allProducts;
57 insert allPbes;
73 insert allLines;
```

## ■ Zoom · Transaction control

```
1 Savepoint sp = Database.setSavepoint();
2 try {
3     // ... lists, stitch, four inserts
4 } catch (Exception e) {
5     Database.rollback(sp);
6     throw new DispatchServiceException(e.getMessage(), e);
7 }
```

# With Unit of Work — Lines: 31 (58% ↓)

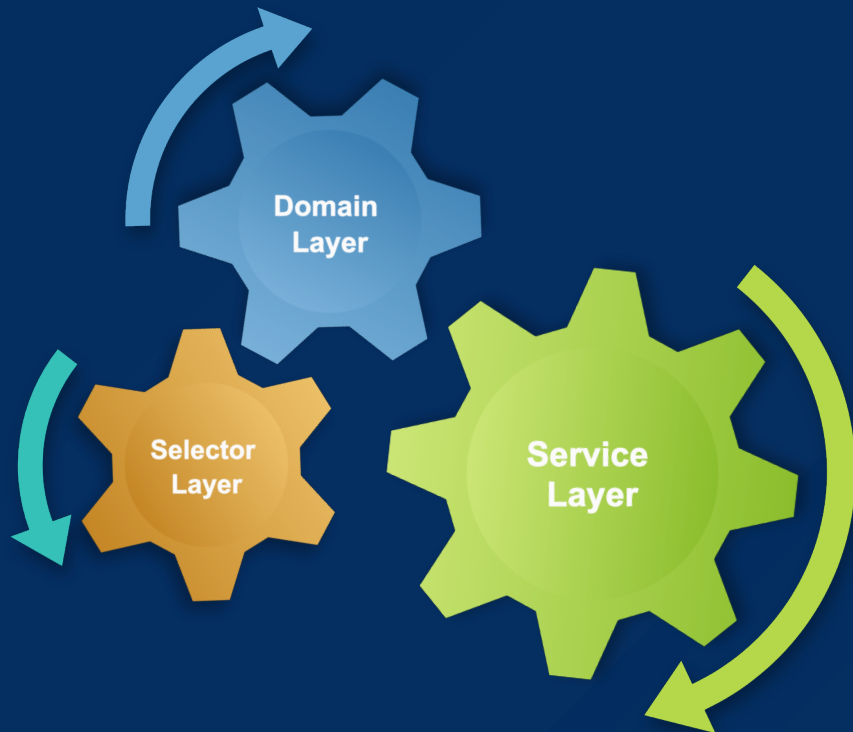
## ■ Same work — register, then commit

```
1 fflib_SObjectUnitOfWork uow = new fflib_SObjectUnitOfWork(  
2     new List<Schema.SObjectType> {  
3         Product2.SObjectType,  
4         PricebookEntry.SObjectType,  
5         Opportunity.SObjectType,  
6         OpportunityLineItem.SObjectType  
7     });  
8 for (Integer o = 0; o < 10; o++) {  
9     Opportunity opp = new Opportunity();  
10    // ... set Name, StageName, CloseDate  
13    uow.registerNew(opp);  
14    for (Integer i = 0; i < o + 1; i++) {  
15        Product2 product = new Product2();  
16        product.Name = opp.Name + ' : Product : ' + i;  
17        uow.registerNew(product);  
18        PricebookEntry pbe = new PricebookEntry();  
19        // ... set UnitPrice, IsActive, Pricebook2Id  
23        uow.registerNew(pbe, PricebookEntry.Product2Id, product);  
24        OpportunityLineItem oli = new OpportunityLineItem();  
25        // ... set Quantity, TotalPrice  
27        uow.registerRelationship(oli, OpportunityLineItem.PricebookEntryId, pbe);  
28        uow.registerNew(oli, OpportunityLineItem.OpportunityId, opp);  
29    }  
30 }  
31 uow.commitWork();
```

# fflib\_SObjectUnitOfWork Methods

- `fflib_SObjectUnitOfWork(List<SObjectType> types)` — dependency order
- `registerNew(record)` · `registerNew(record, field, parent)`
- `registerRelationship(record, field, related)` — stitch before Ids exist
- `registerDirty(record)` · `registerDeleted(record)`
- `commitWork()` — savepoint, bulk DML, rollback, rethrow
- Other methods: `registerWork` , `registerUpsert` , `registerEmail` ...

# Service Layers Explained



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Service Layer Principles



Warehouse Operations App Overview



Warehouse Operations App Code



Warehouse Operations Agent

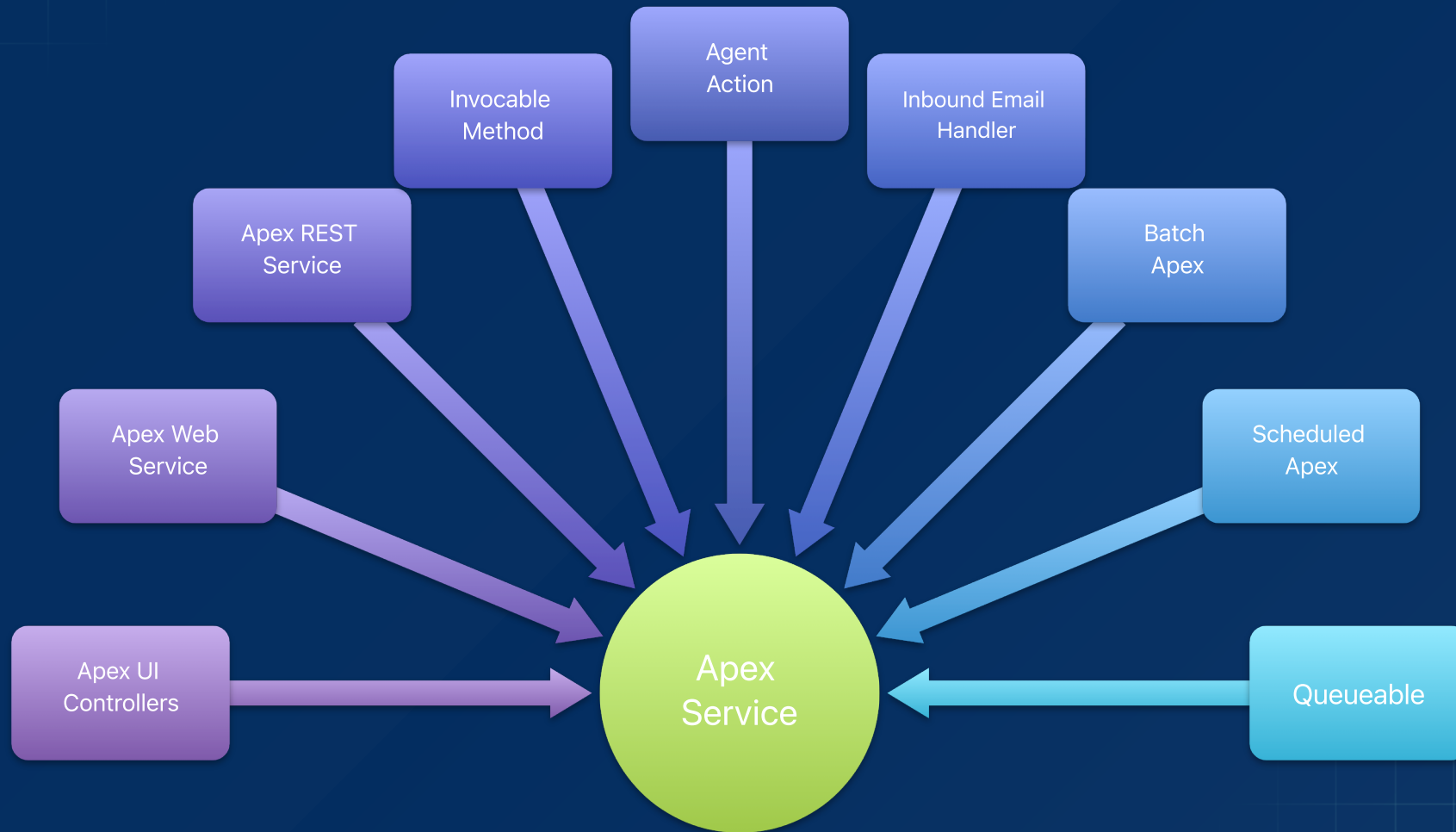


# Service ➔ Task Orientated Logic

- A Service represents a **feature** of the application
  - Fulfillment — `FulfillmentService.cls`
  - Dispatch — `DispatchService.cls`
  - Maintenance — `MaintenanceService.cls`
- Each method is a **task** within that feature — `releaseOrders` ,  
`dispatchWarehouses` , `completeLines`
- Think of it as the **conductor** — the band handles querying and object-specific logic; the Service sets the tempo and the rules of the orchestration



# Services have many Consumers





# Service Checklist

- Client-agnostic inputs and outputs
  -  `dispatchWarehouses(Set<Id> warehouseIds)`
  -  `dispatch(DispatchForm form)`
- Client-agnostic exceptions
  -  `throw DispatchServiceException`
  -  `throw AuraHandledException`
- **Bulkification** — collections in and out; don't force callers to loop
  -  `dispatchWarehouses(Set<Id> warehouseIds)`
  -  `dispatchWarehouse(Id warehouseId)` only
- **Transactions** — one UOW; rollback; Service→Service passes the outer uow
- **Security** — enforce user-mode in DML (UOW) and SOQL (Selector)

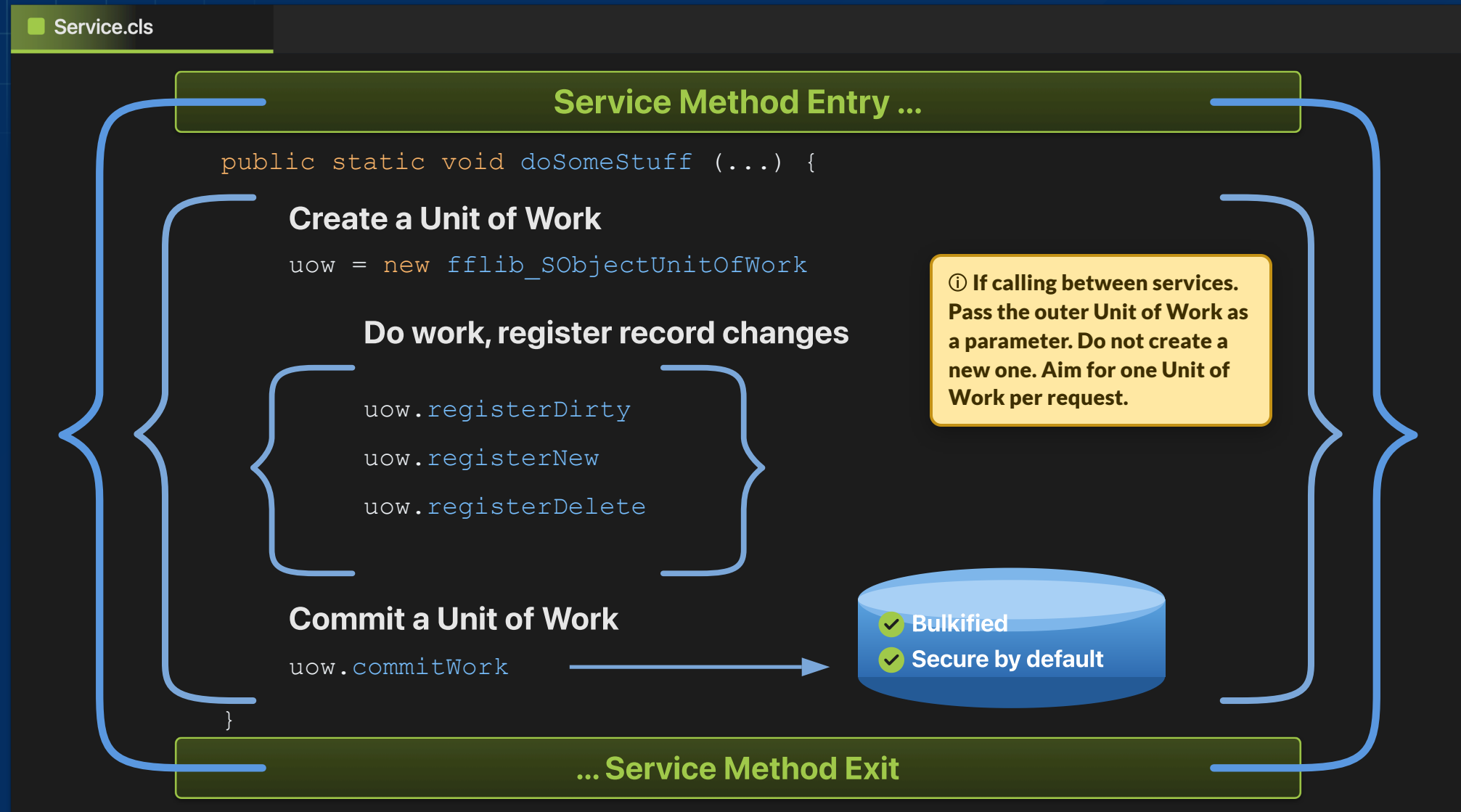
# Service Evolution

- **Services** — now prefer instance methods over static
  -  `DispatchService.newInstance().dispatchWarehouses(...)`
  -  `public static void dispatchWarehouses(...)`
- **Apex interfaces** — now recommended only for dependency injection
  -  `IDispatchService` only when a test or package must inject
  -  `IFulfillmentService` / `IDispatchService` / `IMaintenanceService` on every service by default
- **Application class** — optional; mocking can be done without it — [blog](#)

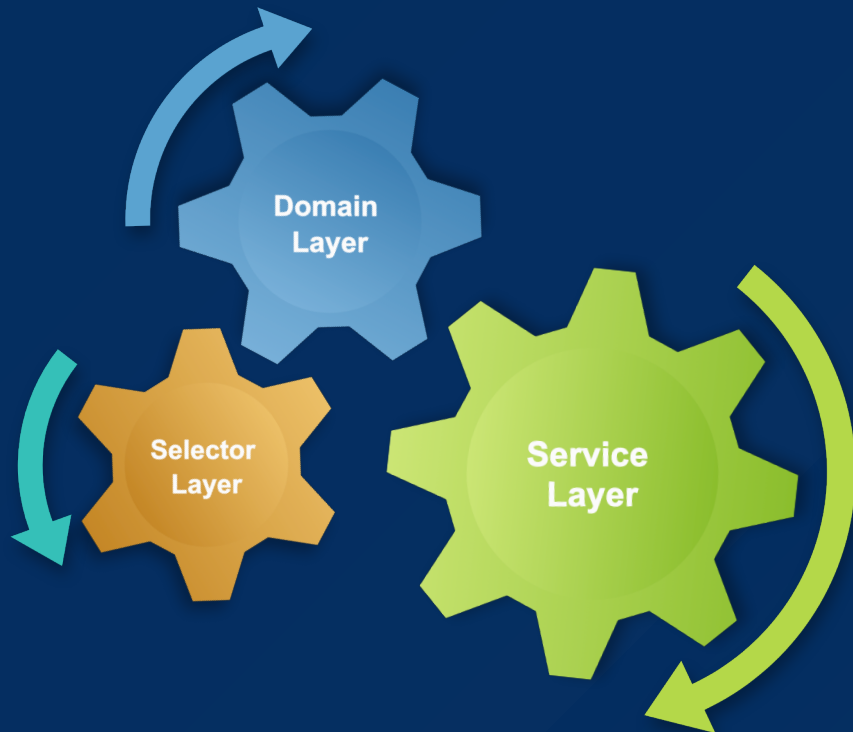
# Service - Consumers and Dependencies



# Service - Leveraging Unit of Work



# Service Layers Explained



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Service Layer Principles



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Warehouse Operations App Code



Warehouse Operations Agent



# Warehouse Operations App

- The operations app for a warehouse that uses robots
- Release pick work, dispatch idle robots, take a worn one into maintenance
- The sample is the **desk**, not the robot brain
- Improve sample clarity between service vs domain
  - `DispatchService` — a process
  - `Robots` — an object
- Leverages multiple clients, LWC, REST API, Batch and AI



# Warehouse App Tabs

Warehouse Operations

HomeWarehouses ▾Robots ▾Robot Models ▾Fulfillment Orders

Warehouse

North Hub

Related

Details

Warehouse Name

North Hub

Status

Active

Location

North

Owner

User User

System Information

Created By

User User, 9/6/2026, 5:29 AM

Last Modified By

User User, 9/6/2026, 5:29 AM

Warehouse Operations

HomeWarehouses ▾Robots ▾Robot Models ▾Fulfillment Orders

Robot

Ember

Related

Details

Robot Name

Ember

Warehouse

North Hub

Model

Picker 100

Status

Idle

Health

Warning

Wear Percent

79.00%

Hours Since Service

39.00

Battery Level

55.00

Owner

Warehouse Operations

HomeWarehouses ▾Robots ▾Robot Models ▾Fulfillment Orders

Fulfillment Order

FO-00000

Related

Details

Fulfillment Order Number

FO-00000

Warehouse

North Hub

Status

Draft

Priority

Normal

Due Date

9/9/2026

Owner

User User

System Information

Warehouse Operations

HomeWarehouses ▾Robots ▾Robot Models ▾Fulfillment Orders

Fulfillment Order

FO-00001

Related

Details

Fulfillment Lines (3)

New

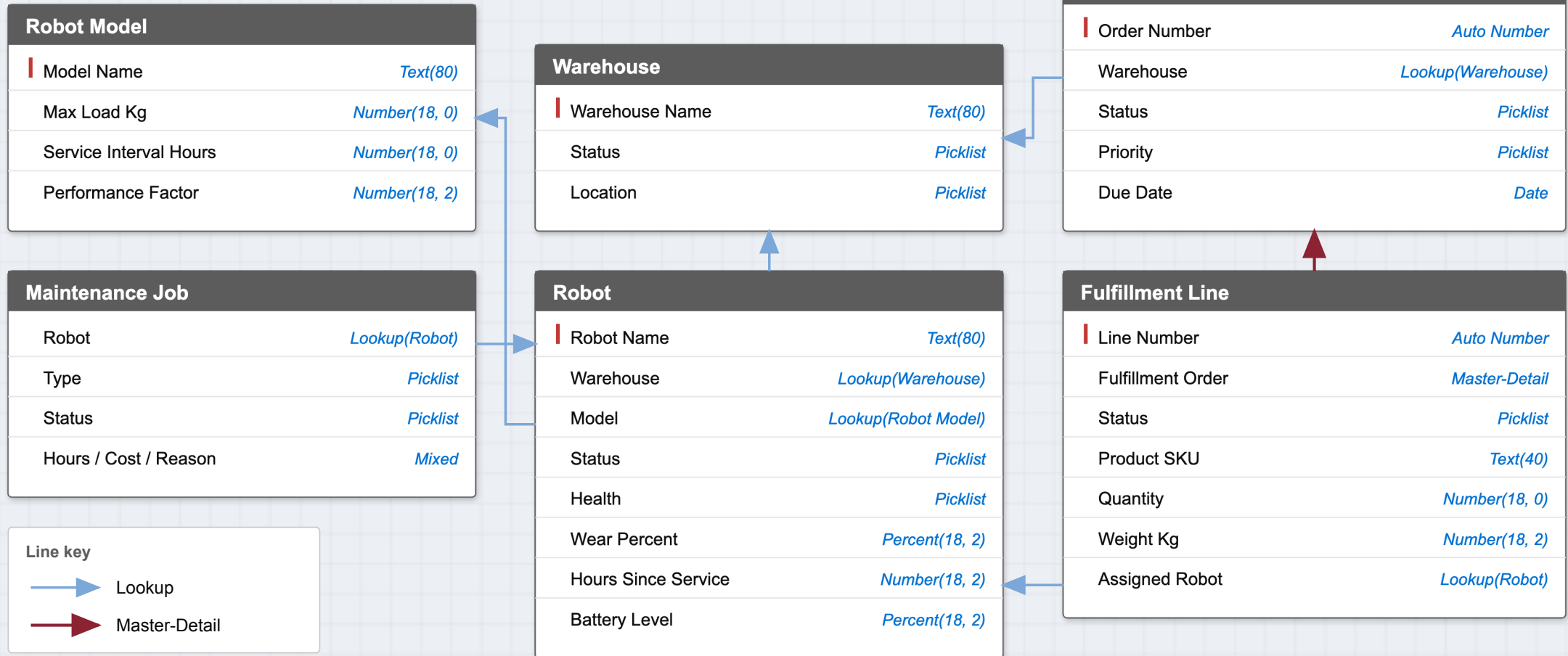
Fulfillment Line Number

FL-00001

FL-00002

FL-00003

# Warehouse App Objects



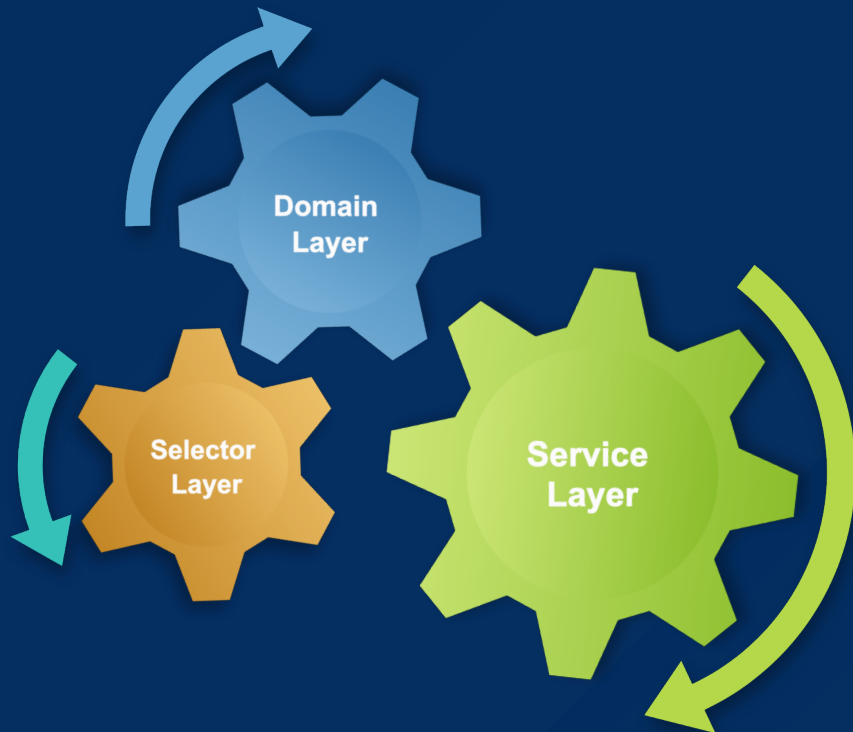


# Warehouse App Processes

## North Hub Warehouse Fulfillment



# Service Layers Explained



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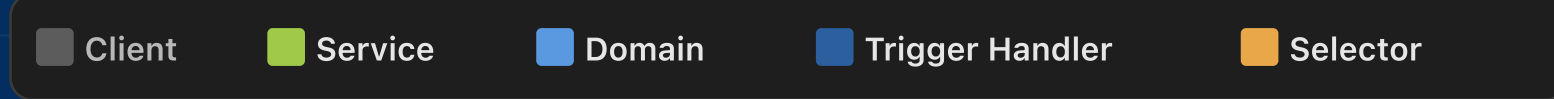
Warehouse Operations App Code



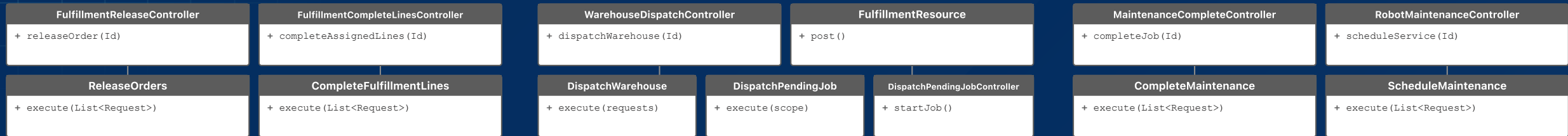
Warehouse Operations Agent



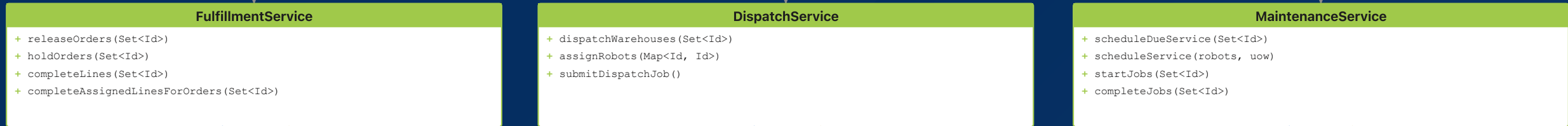
# Warehouse App Classes



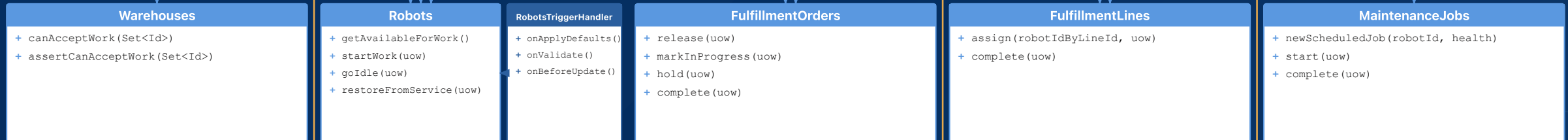
## Clients



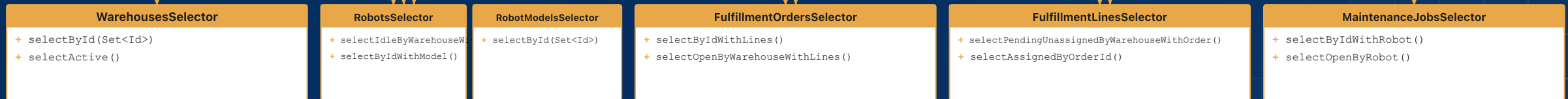
## Services



## Domains



## Selectors



# North Hub Warehouse Fulfillment

## 1 Release FO-00001

Put work on the floor

FulfillmentReleaseController.cls

@AuraEnabled  
releaseOrder (Id)

FulfillmentService.cls

releaseOrders ( Set<Id> )

FulfillmentOrders.cls

release (uow)

FulfillmentOrdersSelector.cls

selectByIdWithLines ( Set<Id> )

## 2 Dispatch North Hub

Put robots on that work

WarehouseDispatchController.cls

@AuraEnabled  
dispatchWarehouse (Id)

DispatchService.cls

dispatchWarehouses ( Set<Id> )

Robots.cls

getAvailableForWork ()

RobotsSelector.cls

selectIdleByWarehouseWithModel ( Set<Id> )

## 3 Complete Lines FO-00001

Finish the picks

FulfillmentCompleteLinesController.cls

@AuraEnabled  
completeAssignedLines (Id)

FulfillmentService.cls

completeAssignedLinesForOrders ( Set<Id> )

MaintenanceService.cls

scheduleService (robots, uow)

Robots.cls

goIdle (uow) applyWear (uow)

## 4 Complete Maintenance The new job

Put Ember back

MaintenanceCompleteController.cls

@AuraEnabled  
completeJob (Id)

MaintenanceService.cls

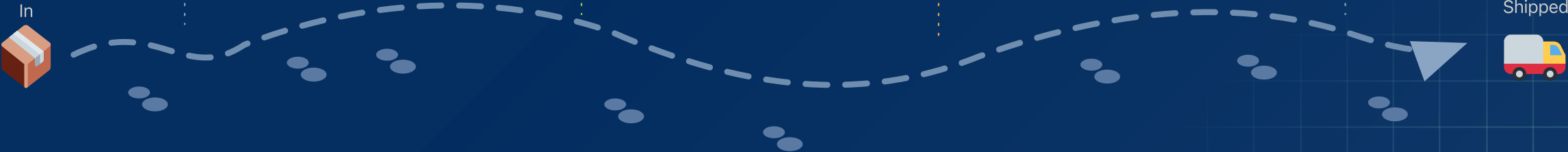
completeJobs ( Set<Id> )

MaintenanceJobs.cls

complete (uow)

Robots.cls

restoreFromService (uow)



WarehouseDispatchController.cls

```
1 public inherited sharing class WarehouseDispatchController {
2     @AuraEnabled(cacheable=false)
3     public static void dispatchWarehouse(Id warehouseId) {
4         try {
5             DispatchService.newInstance()
6                 .dispatchWarehouses(new Set<Id>{ warehouseId });
7         } catch (Exception e) {
8             throw toAura(e);
9         }
10    }
11
12    private static AuraHandledException toAura(Exception e) {
13        AuraHandledException auraException = new AuraHandledException(e.getMessage()
14        auraException.setMessage(e.getMessage());
15        return auraException;
16    }
17 }
```

# DispatchPendingJob.cls

```
1 public inherited sharing class DispatchPendingJob
2     implements System.Schedulable, Database.Batchable<SObject>, Database.Stateful {
3
4     public Database.QueryLocator start(Database.BatchableContext context) {
5         return WarehousesSelector.newInstance()
6             .selectActiveAsQueryLocator();
7     }
8
9     public void execute(
10         Database.BatchableContext context,
11         List<Warehouse__c> warehouses) {
12         try {
13             DispatchService.newInstance()
14                 .dispatchWarehouses(new Map<Id, Warehouse__c>(warehouses).keySet());
15         } catch (Exception e) {
16             jobErrors.add(/* JobError */);
17         }
18     }
19
20     public void finish(Database.BatchableContext context) { /* email JobErrors */
21 }
```

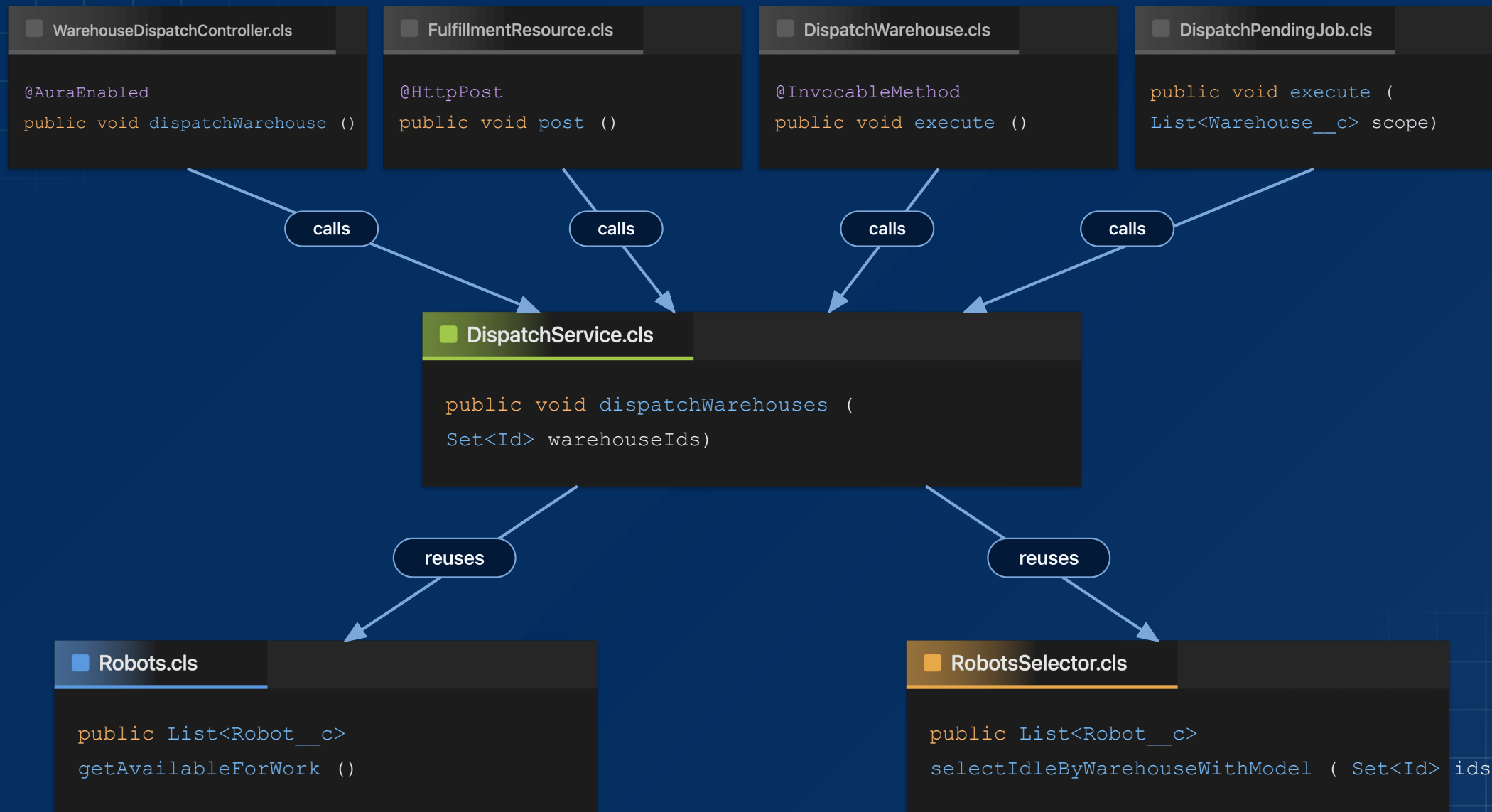
## ■ FulfillmentResource.cls

```
1 @RestResource(UrlMapping='/warehouse/fulfillment/*')
2 global inherited sharing class FulfillmentResource {
3     @HttpPost
4     global static void post() {
5         Request body = (Request) JSON.deserialize(
6             RestContext.request.requestBody.toString(), Request.class);
7         String action = /* last URI segment */;
8         switch on action {
9             when 'release' {
10                 FulfillmentService.newInstance()
11                     .releaseOrders(new Set<Id>(body.orderIds));
12             }
13             when 'dispatch' {
14                 DispatchService.newInstance()
15                     .dispatchWarehouses(new Set<Id>(body.warehouseIds));
16             }
17             when 'complete' {
18                 FulfillmentService.newInstance()
19                     .completeLines(new Set<Id>(body.lineIds));
20             }
21         }
22     }
```

## DispatchWarehouse.cls

```
1 public inherited sharing class DispatchWarehouse {
2     @InvocableMethod(
3         label='Dispatch Warehouse'
4         category='Warehouse Operations')
5     public static List<Result> execute(List<Request> requests) {
6         Set<Id> warehouseIds = new Set<Id>();
7         for (Request request : requests) {
8             warehouseIds.add(request.warehouseId);
9         }
10        DispatchService.newInstance().dispatchWarehouses(warehouseIds);
11        List<Result> results = new List<Result>();
12        for (Request request : requests) {
13            Result result = new Result();
14            result.warehouseId = request.warehouseId;
15            results.add(result);
16        }
17        return results;
18    }
19 }
```





## ■ DispatchService.dispatchWarehouses

```
public virtual void dispatchWarehouses (Set<Id> warehouseIds) {
```

1. **Service** — new `fflib_SObjectUnitOfWork`
2. **Selector** — pending unassigned lines at the warehouses
3. **Selector** — idle robots, with model max load
4. **Domain** — `Robots.getAvailableForWork()` (Idle, not Critical, wear < 90, battery > 20)
5. **Service** — `match` (per warehouse, line weight vs robot max load)
6. **Domain** — `Robots.startWork(uow)` · `FulfillmentLines.assign(uow)` · `FulfillmentOrders.markInProgress(uow)`
7. **Service** — `uow.commitWork()`

```
}
```

## ■ DispatchService.dispatchWarehouses

```
1 public virtual void dispatchWarehouses(Set<Id> warehouseIds) {
2     fflib_SObjectUnitOfWork uow = UnitOfWork.newInstance();
3
4     FulfillmentLinesSelector lineSelector =
5         FulfillmentLinesSelector.newInstance();
6     RobotsSelector robotSelector =
7         RobotsSelector.newInstance();
8     List<FulfillmentLine__c> pending =
9         lineSelector.selectPendingUnassignedByWarehouseWithOrder(warehouseIds);
10    List<Robot__c> idle =
11        robotSelector.selectIdleByWarehouseWithModel(warehouseIds);
12
13    List<Robot__c> available =
14        Robots.newInstance(idle).getAvailableForWork();
15    Map<Id, Id> robotIdByLineId = match(pending, available);
16    Robots.newInstance(robotsToStart).startWork(uow);
17    FulfillmentLines.newInstance(linesToAssign).assign(robotIdByLineId, uow);
18    FulfillmentOrders.newInstance(released).markInProgress(uow);
19
20    uow.commitWork();
21 }
```

## ■ UnitOfWork.cls

```
1 public static fflib_SObjectUnitOfWork newInstance() {  
2     return new fflib_SObjectUnitOfWork(  
3         new List<SObjectType>{  
4             Warehouse__c.SObjectType,  
5             RobotModel__c.SObjectType,  
6             Robot__c.SObjectType,  
7             FulfillmentOrder__c.SObjectType,  
8             FulfillmentLine__c.SObjectType,  
9             MaintenanceJob__c.SObjectType  
10        },  
11        new fflib_SObjectUnitOfWork.UserModeDML()  
12    );  
13 }
```

## ■ FulfillmentService.completeLines

```
public virtual void completeLines (Set<Id> lineIds) {
```

1. **Service** — new `fflib_SObjectUnitOfWork`

2. **Selector** — lines by Id ( `lineIds` )

3. **Domain** — `FulfillmentLines.complete(uow)` · hours by assigned robot

4. **Selector** — robots with model ( `hoursByRobotId` )

5. **Domain** — `Robots.goIdle(uow)` · `applyWear(uow)`

6. **Selector** — parent orders with every sibling line

7. **Domain** — `FulfillmentOrders.complete(uow, lineIds)` · `Robots.getDueForService()`

8. **Service** — `MaintenanceService.scheduleService(dueRobots, uow)`

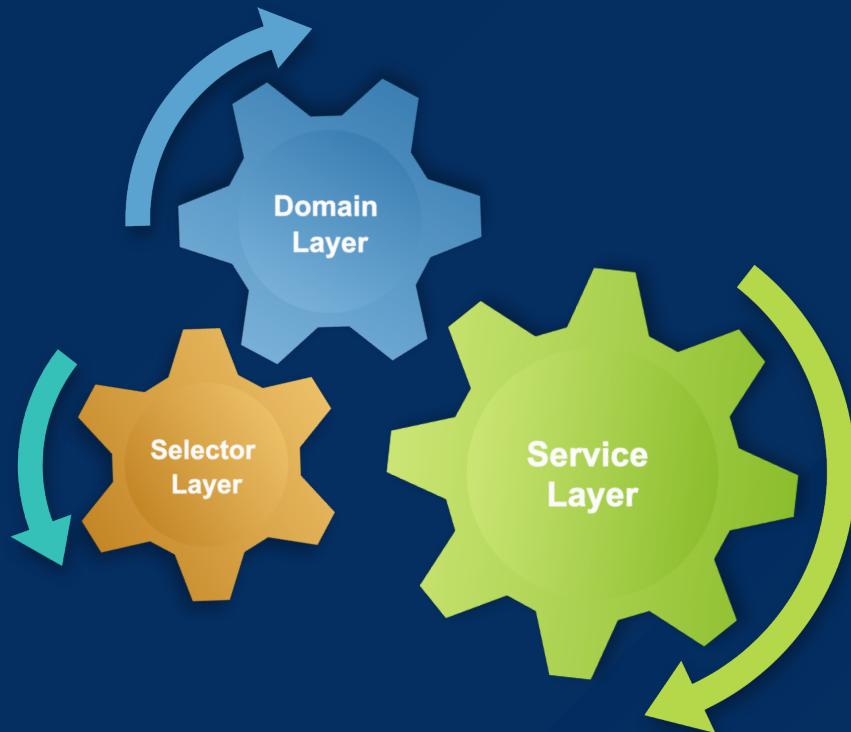
9. **Service** — `uow.commitWork()`

```
}
```

## ■ FulfillmentService.completeLines

```
1 public virtual void completeLines(List<FulfillmentLine__c> lines) {
2     fflib_SObjectUnitOfWork uow = UnitOfWork.newInstance();
3
4     FulfillmentLines fulfillmentLines = FulfillmentLines.newInstance(lines);
5     fulfillmentLines.complete(uow);
6     Map<Id, Decimal> hoursByRobotId =
7         fulfillmentLines.getEstimatedHoursByAssignedRobotId();
8     List<Robot__c> robotRecords =
9         robotSelector.selectByIdWithModel(hoursByRobotId.keySet());
10    Robots robots = Robots.newInstance(robotRecords);
11    robots.goIdle(uow); robots.applyWear(hoursByRobotId, uow);
12
13    List<FulfillmentOrder__c> orders = orderSelector.selectByIdWithLines(orderIds);
14    FulfillmentOrders.newInstance(orders).complete(uow, lineIds);
15
16    Set<Robot__c> dueRobots = robots.getDueForService();
17    if (!dueRobots.isEmpty()) {
18        MaintenanceService.newInstance().scheduleService(dueRobots, uow); // same
19    }
20    uow.commitWork();
21 }
```

# Service Layers Explained



## SERVICE LAYERS EXPLAINED

Separation of Concerns Recap



The Unit of Work



Service Layer Principles



Warehouse Operations App Overview



Warehouse Operations App Code



Warehouse Operations Agent



# Warehouse Agent Actions



## Warehouse Fulfillment

### Actions

■ GetWarehouseDesk.cls

■ CreateRobot (Flow)

■ ReleaseOrders.cls

■ DispatchWarehouse.cls

■ CompleteFulfillmentLines.cls

■ CompleteMaintenance.cls

### Services

→ ■ WarehouseService.cls

→ ■ FulfillmentService.cls

→ ■ DispatchService.cls

→ ■ FulfillmentService.cls

→ ■ MaintenanceService.cls



# Warehouse Agent



Search...



Warehouse Operations

HomeWarehousesRobotsRobot ModelsMore

Robots

 All 

NewImport

3 items • Sorted by Robot Name • Updated a few seconds ago

| <input type="checkbox"/> | Robot Name ↑                   | Warehouse | Status | He |
|--------------------------|--------------------------------|-----------|--------|----|
| 1                        | <input type="checkbox"/> Atlas | North Hub | Idle   | He |
| 2                        | <input type="checkbox"/> Bolt  | North Hub | Idle   | Wa |
| 3                        | <input type="checkbox"/> Ember | North Hub | Idle   | Wa |

Warehouse Fulfillment



Let's chat!

I can review warehouse robots against pending fulfillment work, create robots when you are short, release draft orders, and run one dispatch. What would you like to do?

Describe your task or ask a question...

# Warehouse Agent deploying Robots!





**Warehouse Operations**

[Home](#) [Warehouses](#) [Robots](#) [Robot Models](#) [More](#)

Robots

 **All** 

11 items • Sorted by Robot Name • Updated a few seconds ago

|   | <input type="checkbox"/> Robot Name ↑  | Warehouse | Status  |
|---|----------------------------------------|-----------|---------|
| 1 | <input type="checkbox"/> Atlas         | North Hub | Working |
| 2 | <input type="checkbox"/> Bolt          | North Hub | Working |
| 3 | <input type="checkbox"/> Ember         | North Hub | Working |
| 4 | <input type="checkbox"/> North Spare 1 | North Hub | Working |
| 5 | <input type="checkbox"/> North Spare 2 | North Hub | Working |
| 6 | <input type="checkbox"/> North Spare 3 | North Hub | Working |
| 7 | <input type="checkbox"/> North Spare 4 | North Hub | Working |
| 8 | <input type="checkbox"/> North Spare 5 | North Hub | Working |

**Warehouse Fulfillment** 



I can review warehouse robots against pending fulfillment work, create robots when you are short, release draft orders, and run one dispatch. What would you like to do?



Process North Hub orders optimally with one dispatch and create robots if necessary. Do this now. Do not ask for confirmation.



North Hub has been processed optimally:

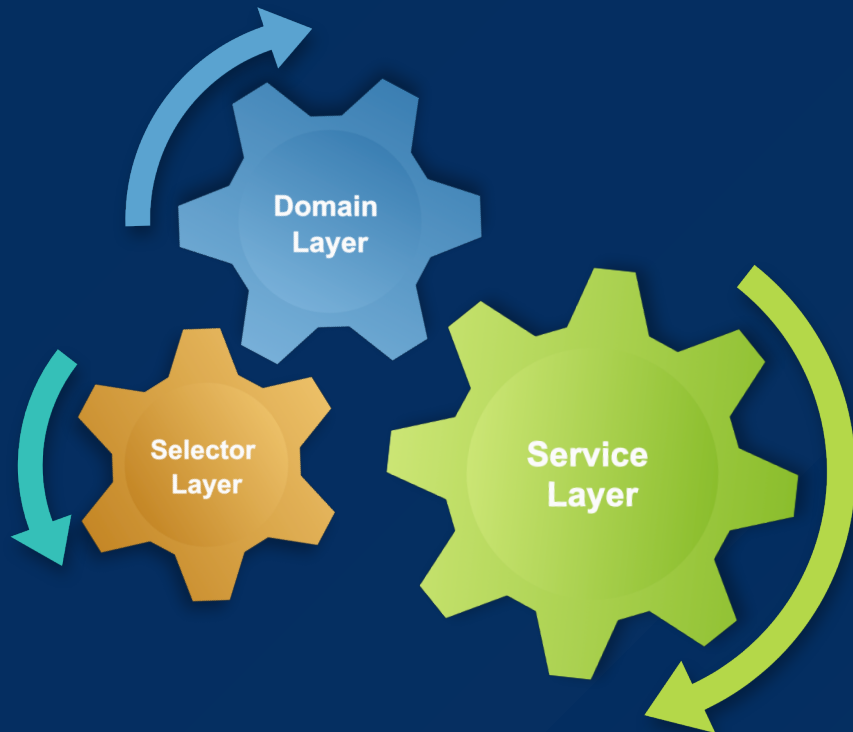
- Created 8 robots: North Spare 1 through North Spare 8.
- Released draft orders: FO-00017, FO-00018, FO-00019, and FO-00020.
- Ran one dispatch at North Hub.

Would you like to process another warehouse or review the results?



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# Service Layers Explained



## SERVICE LAYERS EXPLAINED

Separation of Concerns Recap



The Unit of Work



Service Layer Principles



Warehouse Operations App Overview



Warehouse Operations App Code



Warehouse Operations Agent



# What's next in the series

## SESSION



Session #1 - Separation of Concerns in Apex: Why Your Future Self Will Thank You



Session #2 - Service Layers Explained: Coordinating Business Logic in Apex



Session #3 - Domain vs Service: Where Should Your Apex Logic Live?



Session #4 - Query Logic as a First-Class Architecture Concern



Session #5 - Mocking in Apex: Why It Changes Everything



Session #6 - Enterprise-Scale Apex Across Multiple Packages

# Thank you!

Andrew Fawcett · Code With Sally

**FFLib Series · Session 002**

